Supplemental material

Supplemental Table 1. Systematic literature review search terms and strategy.

Search terms for PubMed (n=1,124)
#1 (((((((((commuting[Title/Abstract] OR active travel[Title/Abstract]) OR active transportation[Title/Abstract]) OR active transport[Title/Abstract]) OR active commuting[Title/Abstract]) OR commuter[Title/Abstract]) OR walking to work[Title/Abstract]) OR ((walking[MeSH] OR walking[All Fields]) AND for transportation[Title/Abstract])) OR cycling to work[Title/Abstract]) OR (cycling[All Fields] AND for transportation[Title/Abstract]) OR ((bicycling[MeSH] OR bicycling[Title/Abstract]) AND for transportation[Title/Abstract]) OR (biking[Title/Abstract] AND for transportation[Title/Abstract]) OR (commuter cycling[Title/Abstract] AND for transportation[Title/Abstract]))
#2 (Obesity[Mesh] OR Overweight[Mesh] OR Body Mass Index[Mesh] OR Waist Circumference[Mesh] OR obesity[Title/Abstract] OR overweight[Title/Abstract] OR fat[Title/Abstract] OR body mass index[Title/Abstract] OR BMI[Title/Abstract] OR adiposity[Title/Abstract] OR waist circumference[Title/Abstract] OR WC[Title/Abstract])
#3 (hypertension[Mesh] OR blood pressure[Mesh] OR hypertension[Title/Abstract] OR blood pressure[Title/Abstract])
#4 (diabetes mellitus[Mesh] OR diabetes[Title/Abstract] OR diabetic[Title/Abstract] OR diabetics[Title/Abstract] OR DM[Title/Abstract] OR T2D[Title/Abstract] OR T2DM[Title/Abstract] OR NIDDM[Title/Abstract])
#5 (Metabolic Syndrome[Mesh] OR metabolic syndrome[Title/Abstract] OR syndrome[Title/Abstract] OR insulin resistance syndrome[Title/Abstract] OR MetS[Title/Abstract])
#6 (#2 OR #3 OR #4 OR #5)
#7 (#1 AND #6)
Search terms for Embase (n=2,608)
#1 commuting.mp. or active travel.mp. or active transportation.mp. or active

transport.mp. or active transport/ or active commuting.mp. or commuter.mp. or walking to work.mp. or walking for transportation.mp. or cycling to work.mp. or cycling for transportation.mp. or bicycling for transportation.mp. or biking for transportation.mp. or commuter cycling for transportation.mp.
#2 obesity.mp. or obesity/ or waist circumference.mp. or waist circumference/ or obes*.mp. or fat*.mp. or overweight*.mp. or adiposit*.mp. or body mass index.mp. or body mass/ or BMI.mp. or WC.mp.
#3 hypertension.mp. or hypertension/ or blood pressure.mp. or blood pressure/
#4 diabetes mellitus.mp. or diabetes mellitus/ or diabetes.mp. or diabetes mellitus/ or diabetic.mp. or diabetics.mp. or DM.mp. or T2D.mp. or T2DM.mp. or non insulin dependent diabetes mellitus/ or NIDDM.mp.
#5 Metabolic Syndrome.mp. or metabolic syndrome X/ or syndrome.mp. or syndrome/ or insulin resistance syndrome.mp. or MetS.mp.
#6 (#2 OR #3 OR #4 OR #5)
#7 (#1 AND #6)
Search terms for Web of Science (n=3,837)
#1 TS=("commuting" OR "active travel" OR "active transportation" OR "active transport" OR "active commuting" OR "commuter" OR "walking to work" OR "walking for transportation" OR "cycling to work" OR "cycling for transportation" OR "bicycling for transportation" OR "biking for transportation" OR "commuter cycling for transportation")
#2 TS=("obes*" OR "overweight*" OR "fat*" OR "body mass index" OR "BMI" OR "adiposit*" OR "waist circumference" OR "WC")
#3 TS=("hypertension" OR "blood pressure")
#4 TS=("diabetes" OR "diabetic" OR "diabetics" OR "DM" OR "T2D" OR "T2DM" OR "NIDDM")
#5 TS=("Metabolic Syndrome" OR "syndrome" OR "insulin resistance syndrome" OR "MetS")
#6 (#2 OR #3 OR #4 OR #5)

#7 (#1 AND #6)
Search terms for Open Grey (n=1)
#1 "commuting" OR "active travel" OR "active transportation" OR "active transport" OR "active commuting" OR "commuter" OR "walking to work" OR "walking for transportation" OR "cycling to work" OR "cycling for transportation" OR "bicycling for transportation" OR "biking for transportation" OR "commuter cycling for transportation"
#2 "obes*" OR "overweight*" OR "fat*" OR "body mass index" OR "BMI" OR "adiposit*" OR "waist circumference" OR "WC"
#3 "hypertension" OR "blood pressure"
#4 "diabetes" OR "diabetic" OR "diabetics" OR "DM" OR "T2D" OR "T2DM" OR "NIDDM"
#5 "Metabolic Syndrome" OR "syndrome" OR "insulin resistance syndrome" OR "MetS"
#6 (#2 OR #3 OR #4 OR #5)
#7 (#1 AND #6)

Supplemental Table 2. Characteristics of included studies for prehypertension prevalence.

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
Berglund	2016	Sweden	NA	cross-sectional	45–75	Men&Women	Walking & bicycling to work	self-reported	BMI≥25 kg/m²	Traveling mode, gender, age, education level, occupation, smoking habits, choosing food for health reasons
Baldew	2017	Asian-Suriname	the Healthy Life in Suriname (HELISUR) study.	cross-sectional	18–70	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Sex, age and for PA in other domains
Baldew	2017	African-Suriname	the Healthy Life in Suriname (HELISUR) study.	cross-sectional	18–70	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Sex, age and for PA in other domains
Dąbrowska	2015	Poland	NA	cross-sectional	45-55	Women	Active transportation	self-reported	BMI≥30 kg/m²	None
Gordon-Larsen	2009	America	the Coronary Artery Risk Development in Young	cross-sectional	45	Men	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Age, race, income, education, and examination center, smoking, alcohol consumption, physical activity index excluding walking

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Adults (CARDIA) study							
Gordon-Larsen	2009	America	the Coronary Artery Risk Development in Young Adults (CARDIA) study	cross-sectional	45	Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Age, race, income, education, and examination center, smoking, alcohol consumption, physical activity index excluding walking
Grøntved	2016	Sweden	the V€asterbottens Health Survey	cohort study	43.5	Men&Women	Walking or bicycling to work	self-reported	BMI≥30 kg/m²	Age at baseline, follow-up time, baseline levels of risk facto, and sex, leisure time exercise, occupational physical activity, smoking status, educational status, alcohol consumption, and intake of coffee, total energy, fruit, vegetables, trans fat, and fiber
Hu	2001	China	World	cross-sectional	15–69	Men	Walking &	self-reported	BMI≥25 kg/m²	Age, education, smoking,

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Bank				bicycling to work			alcohol consumption, body mass index (except with overweight as dependent variable) and occupational activity
Hu	2001	China	World Bank	cross-sectional	15–69	Women	Walking & bicycling to work	self-reported	BMI≥25 kg/m²	Age, education, smoking, alcohol consumption, body mass index (except with overweight as dependent variable) and occupational activity
Huang	2017	Taiwan	NA	cross-sectional	43.0 ± 9.9	Men&Women	Walking & bicycling to work	self-reported	WC (cm)≥90 for men or≥80 for women	Gender, age and nutritional health behavior
Kwaśniewska	2010	Poland	the National Multicentre Health Survey WOBASZ, Poland (2002–2005)	cross-sectional	20–74	Men	Walking & bicycling to work	self-reported	WC≥102 cm	Age, education, place of residence, monthly income, smoking, alcohol consumption, calorie intake, LTPA and BMI
Kwaśniewska	2010	Poland	the	cross-sectional	20–74	Women	Walking &	self-reported	WC≥88 cm	Age, education, place of

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			National Multicentre Health Survey WOBASZ, Poland (2002–2005)				bicycling to work			residence, monthly income, smoking, alcohol consumption, calorie intake, LTPA and BMI
Laverty	2015	China	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	India	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
Laverty	2015	Mexico	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	Ghana	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	Russia	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
Laverty	2015	South Africa	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2013	United Kingdom	a nationally representative survey of UK residents in 2009/2011	cross-sectional	16-65	Men&Women	Walking or bicycling to work	self-reported	BMI≥30 kg/m²	Age, gender, ethnic group, education level, NS-SEC Social Class, and place of residence
Lerssrimongkol	2016	Thailand	The cross-sectional Chiang Mai University (CMU) Health Worker Study	cross-sectional	40	Men&Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Age, sex, marital status, educational level, income, occupation, smoking, health risk due to alcohol consumption, physical activity level and BMI

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
Lerssrimongkol	2016	Thailand	The cross-sectional Chiang Mai University (CMU) Health Worker Study	cross-sectional	40	Men&Women	Walking & bicycling to work	self-reported	WC≥90 cm in men and≥80 cm in women	Age, sex, marital status, educational level, income, occupation, smoking, health risk due to alcohol consumption, physical activity level and BMI
Lindström	2008	Sweden	the cross-sectional 2004 public health survey in Skåne	cross-sectional	18–80	Men	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Age, country of origin, education and time for travel to work (one way)
Lindström	2008	Sweden	the cross-sectional 2004 public health survey in Skåne	cross-sectional	18–80	Women	Walking & bicycling to work	self-reported	BMI≥30 kg/m²	Age, country of origin, education and time for travel to work (one way)
Lu	2013	China	NA	cross-sectional	18–94	Men&Women	Walking,	self-reported	BMI≥24 kg/m²	Age, gender, educational level,

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
							cycling, and using public transport			household income level, and overall physical activity level
Lu	2013	China	NA	cross-sectional	18–94	Men&Women	Walking, cycling, and using public transport	self-reported	WC⩾90 cm for men or⩾80 cm for women	Age, gender, educational level, household income level, and overall physical activity level
McKay	2015	India and Bangladesh	the Chronic Disease Risk Factor Survey	cross-sectional	≥18	Men&Women	Active travel	self-reported	WC>85 cm for men, >80 cm for women	Age, sex, site, education, smoking status, oil/butter consumption, and work- and leisure-related physical activity
Millett	2013	Indian	the Indian Migration Study	cross-sectional	42.2±9.6	Men&Women	Walking or bicycling to work	self-reported	BMI≥30 kg/m²	Age, sex, caste, standard of living, occupation, factory location, leisure time physical activity, fat intake, smoking status, alcohol intake
Omar	2008	Europe		cross-sectional	≥15	Men	Commiting PA	self-reported	BMI≥30 kg/m²	Age, education, having a chronic illness, and dietary habits
Omar	2008	Europe		cross-sectional	≥15	Women	Commiting PA	self-reported	BMI≥30 kg/m²	Age, education, having a chronic illness, and dietary habits
Pearson	2014	New	the 2006/7	cross-sectional	≥15	Men&Women	Active travel	self-reported	BMI≥30 kg/m²	Age, sex, ethnicity, Economic

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
		Zealand	New Zealand Health Survey							Living Standard Index, individual-level deprivation, highest educational qualification, household composition, smoking status and alcohol consumption.
Wen	2007	Australia	the 2003 New South Wales Adult Health Survey	cross-sectional	≥16	Men	Walking or bicycling to work	self-reported	BMI≥30 kg/m²	Level of physical activity, age, marital status, education, main language spoken at home
Wen	2007	Australia	the 2003 New South Wales Adult Health Survey	cross-sectional	≥16	Women	Walking to work	self-reported	BMI≥30 kg/m²	Level of physical activity, age, marital status, education, main language spoken at home
Xiao	2016	China	Nantong Metabolic Syndrome Study	cross-sectional	18–74	Men	Walking or bicycling to work	self-reported	WC≥85 cm	Age at interview, BMI, personal income, tea consumption, drinking status, and intake of red meat, white meat, fish, fruits, and soy food and mutually adjusted for

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
										leisure-time PA, occupational PA, and commuting PA categories
Xiao	2016	China	Nantong Metabolic Syndrome Study	cross-sectional	18–74	Women	Walking or bicycling to work	self-reported	WC≥85 cm	Age at interview, BMI (not included in waist circumference calculation), personal income, tea consumption, drinking status, and intake of red meat, white meat, fish, fruits, and soy food and mutually adjusted for leisure-time PA, occupational PA, and commuting PA categories
Xu	2017	China	NA	cross-sectional	NA	Men	Walking to work	self-reported	BMI ≥ 24 kg/m²	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Men	Bicycling to work	self-reported	BMI ≥ 24 kg/m²	Age, educational level, daily total caloric intake, smoking, drinking, district economic

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
										level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Women	Walking to work	self-reported	BMI ≥ 24 kg/m²	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Women	Bicycling to work	self-reported	BMI ≥ 24 kg/m²	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Men	Walking to work	self-reported	WC ≥85 cm	Aage, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
										transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Men	Bicycling to work	self-reported	WC ≥85 cm	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Women	Walking to work	self-reported	WC ≥80 cm	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Xu	2017	China	NA	cross-sectional	NA	Women	Bicycling to work	self-reported	WC ≥80 cm	Age, educational level, daily total caloric intake, smoking, drinking, district economic level, occupational activity, leisure-time physical activity, leisure-time sitting, and transportation to/from work
Zwald	2018	United States	NHANES	cross-sectional	≥20	Men&Women	Walking & bicycling to	self-reported	BMI≥30 kg/m²	Age, sex, race/Hispanic origin, education, poverty, smoking,

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
							work			survey cycle, non-transportation PA, and urbanization level
Hypertension										
Barengo	2005	Finland	NA	cohort study	25–64	Men & Women	Walking & bicycling to work	self-reported	SBP>160 mmHg or DBP>95 mmHg, in principle. All patients receiving special reimbursement for medication for hypertension are entered into a register maintained by the Social Insurance Institution.	Age, area, time of survey, education, smoking status, alcohol intake, body mass index, baseline systolic blood pressure, and two other physical activitie
Byambasukh	2019	China	The Lifelines study	cross-sectional	45 (36–51)	Men & Women	Walking & bicycling to work	self-reported	SBP≥140 mm Hg and/or DBP≥90 mm Hg and/or use of antihypertensive medication.	Age, square of age, sex, education, smoking, alcohol consumption and BMI
Grøntved	2016	Sweden	the V€asterbot	cohort study	43.5	Men&Women	Walking or bicycling to	self-reported	SBP or DBP exceeded 140 and	Age at baseline, follow-up time, baseline levels of risk

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			tens Health Survey				work		90 mm Hg and/or use of antihypertensive medications	factor, and sex, leisure time exercise, occupational physical activity, smoking status, educational status, alcohol consumption, and intake of coffee, total energy, fruit, vegetables, trans fat, and fiber
Hayashi	1999	Japan	The Osaka Health Survey	cohort study	35-60	Men	Walking to work	self-reported	SBP≥160 mm Hg or/and DBP≥95 mm Hg, or use of antihypertensive medication	Age, body mass index, alcohol consumption, leisure-time physical activity, smoking status, fasting plasma glucose level, systolic blood pressure, and diastolic blood pressure.
Hu	2001	China	World Bank	cross-sectional	15–69	Men	Walking & bicycling to work	self-reported	SBP≥140 mm Hg and/or DBP≥90 mm Hg, or using antihypertension drugs.	Age, education, smoking, alcohol consumption, body mass index and occupational activity
Hu	2001	China	World Bank	cross-sectional	15–69	Women	Walking & bicycling to work	self-reported	SBP≥140 mm Hg and/or DBP≥90 mm Hg, or using antihypertension drugs.	Age, education, smoking, alcohol consumption, body mass index and occupational activity

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
Huang	2017	Taiwan	NA	cross-sectional	43.0 ± 9.9	Men&Women	Walking & bicycling to work	self-reported	SBP (mm Hg)≥130/DBP(mm Hg)≧85	Gender, age and nutritional health behavior
Kwaśniewska	2010	Poland	the National Multicentre Health Survey WOBASZ, Poland (2002–2005)	cross-sectional	20–74	Men	Walking & bicycling to work	self-reported	SBP and/or DBP≥130 and/or≥85 mm Hg	Age, education, place of residence, monthly income, smoking, alcohol consumption, calorie intake, LTPA and BMI
Kwaśniewska	2010	Poland	the National Multicentre Health Survey WOBASZ, Poland (2002–2005)	cross-sectional	20–74	Women	Walking & bicycling to work	self-reported	SBP and/or DBP≥130 and/or≥85 mm Hg	Age, education, place of residence, monthly income, smoking, alcohol consumption, calorie intake, LTPA and BMI
Laverty	2013	United Kingdom	a nationally representat	cross-sectional	16-65	Men&Women	Walking or bicycling to work	self-reported	self-report	Age, gender, ethnic group, education level, NS-SEC Social Class, and place of

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			ive survey of UK residents in 2009/2011							residence.
Lerssrimongkol	2016	Thailand	The cross-sectional Chiang Mai University (CMU) Health Worker Study	cross-sectional	40	Men&Women	Walking & bicycling to work	self-reported	BP ≥140/90 mm Hg or a history of taking anti-hypertensive medication	Age, sex, marital status, educational level, income, occupation, smoking, health risk due to alcohol consumption, physical activity level and BMI
Lu	2013	China	NA	cross-sectional	18–94	Men&Women	Walking, cycling, and using public transport	self-reported	SBP⩾140mmHg and/or DBP⩾90mmHg	Age, gender, educational level, household income level, and overall physical activity level
Millett	2013	Indian	the Indian Migration Study	cross-sectional	42.2±9.6	Men&Women	Walking or bicycling to work	self-reported	doctor diagnosis	Age, sex, caste, standard of living, occupation, factory location, leisure time physical activity, fat intake, smoking status, alcohol intake
Treff	2016	Brazil	Brazilian Longitudin	cross-sectional	35-74	Men	Walking & bicycling to	self-reported	SBP⩾140 mm Hg, DBP⩾90 mm Hg or	Age, race, education, income, education × income, urinary

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			al Study of Adult Health				work		the use of medication to control hypertension	Na excretion, alcohol intake, BMI, urinary Na excretion × BMI.
Treff	2016	Brazil	Brazilian Longitudinal Study of Adult Health	cross-sectional	35-74	Women	Walking & bicycling to work	self-reported	SBP⩾140 mm Hg, DBP⩾90 mm Hg or the use of medication to control hypertension	Age, race, education, income, education × income, urinary Na excretion, alcohol intake, BMI, urinary Na excretion × BMI.
Zwald	2018	United States	NHANES	cross-sectional	≥20	Men&Women	Walking & bicycling to work	self-reported	SBP≥140mmHg, DBP≥90mmHg, or self-reported current use of hypertension medication	Age, sex, race/Hispanic origin, education, poverty, smoking, survey cycle, non-transportation PA, and urbanization level
Diabetes										
Honda	2015	Japan	the Japan Epidemiology Collaboration on Occupational Health (J-ECOH)	cohort study	30-64	Men&Women	Walking to work	self-reported	HbA1c≥6.5 % (48 mmol/mol), fasting plasma glucose≥126 mg/dl (7.0 mmol/l), random plasma glucose≥200 mg/dl (11.1 mmol/l), or	Age, sex, shift work, sleep duration, alcohol consumption, smoking, hypertension, a family history of diabetes, occupational activity, walking for commuting to and from work, and moderateintensity exercise for vigorous-intensity

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Study						currently under medical treatment for diabetes.	exercise and vigorous-intensity exercise for moderate-intensity exercise body mass index
Hu	2003	Finland	NA	cohort study	35-64	Men	Walking & bicycling to work	self-reported	the National Hospital Discharge Register and the National Social Insurance Institution's Register	Age, study year, systolic blood pressure, smoking status, education, other two physical activity and BMI
Hu	2003	Finland	NA	cohort study	35-64	Women	Walking & bicycling to work	self-reported	the National Hospital Discharge Register and the National Social Insurance Institution's Register	Age, study year, systolic blood pressure, smoking status, education, other two physical activity and BMI
Huang	2017	Taiwan	NA	cross-sectional	43.0±9.9	Men&Women	Walking & bicycling to work	self-reported	FBG (mg/dL)≥100	Gender, age and nutritional health behavior
Kwaśniewska	2010	Poland	the National Multicentr	cross-sectional	20–74	Men	Walking & bicycling to work	self-reported	FPG≥5.6mmol/l	Age, education, place of residence, monthly income, smoking, alcohol

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			e Health Survey WOBASZ, Poland (2002–2005)							consumption, calorie intake, LTPA and BMI (in analysis of lipids, BP and FPG)
Kwaśniewska	2010	Poland	the National Multicentre Health Survey WOBASZ, Poland (2002–2005)	cross-sectional	20–74	Women	Walking & bicycling to work	self-reported	FPG≥5.6mmol/l	Age, education, place of residence, monthly income, smoking, alcohol consumption, calorie intake, LTPA and BMI (in analysis of lipids, BP and FPG)
Laverty	2015	China	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	self-reported	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	India	the WHO	cross-sectional	58±14.7	Men&Women	Walking &	self-reported	self-reported	Mean centred age, sex, marital

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Study on Global Ageing and Adult Health (SAGE)				bicycling to work			status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	Mexico	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	self-reported	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	Ghana	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	self-reported	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	Russia	the WHO	cross-sectional	58±14.7	Men&Women	Walking &	self-reported	self-reported	Mean centred age, sex, marital

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Study on Global Ageing and Adult Health (SAGE)				bicycling to work			status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2015	South Africa	the WHO Study on Global Ageing and Adult Health (SAGE)	cross-sectional	58±14.7	Men&Women	Walking & bicycling to work	self-reported	self-reported	Mean centred age, sex, marital status, education, location, household wealth quintile, minutes of vigorous and moderate physical activity, fruit and vegetable consumption, smoking status and alcohol use
Laverty	2013	United Kingdom	a nationally representative survey of UK residents in 2009/2011	cross-sectional	16-65	Men&Women	Walking or bicycling to work	self-reported	self-reported	Age, gender, ethnic group, education level, NS-SEC Social Class, and place of residence.
Lerssrimongkol	2016	Thailand	The cross-sectional	cross-sectional	40	Men&Women	Walking & bicycling to	self-reported	Fasting glucose≥126 mg/dl	Age, sex, marital status, educational level, income,

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
			Chiang Mai University (CMU) Health Worker Study				work		or a history of taking anti-diabetic medication/insulin	occupation, smoking, health risk due to alcohol consumption, physical activity level and BMI
Lu	2013	China	NA	cross-sectional	18–94	Men&Women	Walking, cycling, and using public transport to work	self-reported	FBG⩾7.0mmol/L	Age, gender, educational level, household income level, and overall physical activity level
Millett	2013	Indian	the Indian Migration Study	cross-sectional	42.2±9.6	Men&Women	Walking or bicycling to work	self-reported	doctor diagnosis	Age, sex, caste, standard of living, occupation, factory location, leisure time physical activity, fat intake, smoking status, alcohol intake
Rasmussen	2016	Denmark	NA	cohort study	50–65	Men&Women	Bicycling to work	self-reported	registration in the Danish NDR	Age at the second examination, sex, years of basic school, years of higher education, alcohol intake, dietary energy intake, coffee consumption, polyunsaturated:saturated fat ratio, physical activity at work,

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
										smoking status, paternal diabetes, maternal diabetes, and leisure-time physical activity other than commuter cycling
Sato	2007	Japan	The Kansai Healthcare Study	cohort study	40–55	Men	Walking to work	self-reported	FPG≥126 mg/dl or if participants were taking oral hypoglycemic medication or insulin	Age, fasting plasma glucose level, and daily alcohol consumption showed a nonlinear association with the incidence of type 2 diabetes
Xiao	2016	China	Nantong Metabolic Syndrome Study	Cross-sectional study	18–74	Men	Walking or bicycling to work	self-reported	FPG≥5.6 mmol/L	Age at interview, BMI, personal income, tea consumption, drinking status, and intake of red meat, white meat, fish, fruits, and soy food and mutually adjusted for leisure-time PA, occupational PA, and commuting PA categories
Xiao	2016	China	Nantong Metabolic Syndrome Study	Cross-sectional study	18–74	Women	Walking or bicycling to work	self-reported	FPG≥5.6 mmol/L	Age at interview, BMI, personal income, tea consumption, drinking status, and intake of red meat, white

First author	Published year	Country	Date source	Studies	Mean/median age and age range (years)	Sex	Exposure	Exposure measurement	Outcome definition	Adjustment
										meat, fish, fruits, and soy food and mutually adjusted for leisure-time PA, occupational PA, and commuting PA categories
Zwald	2018	United States	NHANES	cross-sectional	≥20	Men&Women	Walking & bicycling to work	self-reported	measured	Age, sex, race/Hispanic origin, education, poverty, smoking, survey cycle, non-transportation PA, and urbanization level

Supplemental Table 3. Assessment of quality of included cross-sectional studies (Agency for Healthcare Research and Quality [AHRQ]).

First author (year)	Item											Total
	a	b	c	d	e	f	g	h	i	j	k	
Obesity												
Baldew (2017)	1	1	1	1	1	1	1	1	1	1	1	11
Berglund (2016)	1	1	1	1	1	0	1	1	1	1	1	10
Dąbrowska (2015)	1	1	1	1	1	0	1	0	1	0	1	8
Gordon-Larsen (2009)	1	1	1	1	1	1	1	1	1	1	1	11
Hu (2001)	1	1	1	1	1	1	1	1	1	1	1	11
Huang (2017)	1	1	1	1	1	1	1	1	1	0	1	10
Kwaśniewska (2010)	1	1	1	1	1	1	1	1	1	1	1	11
Laverty (2013)	1	1	1	1	1	0	1	1	1	0	1	9
Laverty (2015)	1	1	1	1	1	1	1	1	1	1	1	11
Lerssrimongkol (2016)	1	1	1	1	1	1	1	1	1	0	1	10
Lindström (2008)	1	1	1	1	1	0	1	1	0	1	1	9
Lu (2013)	1	1	1	1	1	1	1	1	1	1	1	11
Millett (2013)	1	1	1	1	1	1	1	1	1	1	1	11
Omar (2008)	1	1	1	1	1	0	1	1	1	1	1	10
Pearson (2014)	1	1	1	1	1	1	1	1	1	1	1	11
Wen (2007)	1	1	1	1	1	0	1	1	1	1	1	10
Xiao (2016)	1	1	1	1	1	1	1	1	1	1	1	11
Xu (2017)	1	1	1	1	1	1	1	1	1	1	1	11
Zwald (2018)	1	1	1	1	1	1	1	1	1	1	1	11
Hypertension												
Byambasukh (2019)	1	1	1	1	1	1	1	1	1	1	1	11
Hayashi (1999)	1	1	1	1	1	1	1	1	1	1	1	11
Hu (2001)	1	1	1	1	1	1	1	1	1	1	1	11
Huang (2017)	1	1	1	1	1	1	1	1	1	0	1	10
Kwaśniewska (2010)	1	1	1	1	1	1	1	1	1	1	1	11
Laverty (2013)	1	1	1	1	1	0	1	1	1	0	1	9
Lerssrimongkol (2016)	1	1	1	1	1	1	1	1	1	0	1	10
Lu (2013)	1	1	1	1	1	1	1	1	1	1	1	11
Millett (2013)	1	1	1	1	1	1	1	1	1	1	1	11
Treff (2016)	1	1	1	1	1	1	1	1	1	1	1	11
Zwald (2018)	1	1	1	1	1	1	1	1	1	1	1	11
Diabetes												
Huang (2017)	1	1	1	1	1	1	1	1	1	0	1	10
Kwaśniewska (2010)	1	1	1	1	1	1	1	1	1	1	1	11
Laverty (2013)	1	1	1	1	1	0	1	1	1	0	1	10
Laverty (2015)	1	1	1	1	1	0	1	1	1	1	1	10
Lerssrimongkol (2016)	1	1	1	1	1	1	1	1	1	0	1	10
Lu (2013)	1	1	1	1	1	1	1	1	1	1	1	11
Millett (2013)	1	1	1	1	1	1	1	1	1	1	1	11

Xiao (2016)	1	1	1	1	1	1	1	1	1	1	1	11
Zwald (2018)	1	1	1	1	1	1	1	1	1	1	1	11

Define the source of information (survey, record review);

b. List inclusion and exclusion criteria for exposed and unexposed subjects (cases and controls) or refer to previous publications;

c. Indicate time period used for identifying patients;

d. Indicate whether or not subjects were consecutive if not population-based;

e. Indicate if evaluators of subjective components of study were masked to other aspects of the status of the participants;

f. Describe any assessments undertaken for quality assurance purposes (e.g., test/retest of primary outcome measurements);

g. Explain any patient exclusions from analysis;

h. Describe how confounding was assessed and/or controlled;

i. If applicable, explain how missing data were handled in the analysis;

j. Summarize patient response rates and completeness of data collection;

k. Clarify what follow-up, if any, was expected and the percentage of patients for which incomplete data or follow-up was obtained.

Supplemental Table 4. Assessment of quality of included cohort studies (NOS).

First author (year)	Study Selection				Comparability of cohorts		Outcome			Total
	a	b	c	d	e	f	g	h	i	
Obesity										
Grøntved (2016)	1	1	1	1	1	1	1	1	1	9
Mytton (2016)	1	1	1	1	1	1	0	0	1	7
Hypertension										
Barengo (2005)	1	1	1	1	1	1	1	1	1	9
Grøntved (2016)	1	1	1	1	1	1	1	1	1	9
Diabetes										
Honda (2015)	0	1	1	1	1	1	1	1	1	8
Hu (2003)	0	1	1	1	1	1	1	1	1	8
Rasmussen (2016)	1	1	1	1	1	1	1	1	1	9
Sato (2007)	1	1	1	1	1	1	1	0	1	8

a. Representativeness of the exposed cohort;

b. Selection of the non-exposed cohort;

c. Ascertainment of exposure;

d. Demonstration that outcome of interest was not present at start of study;

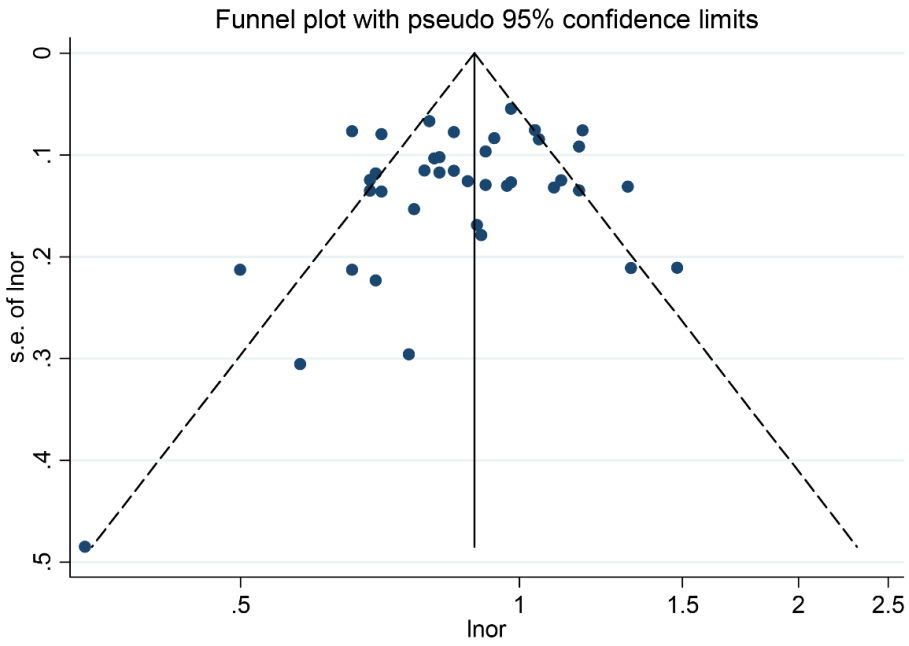
e. Comparability of cohorts on the basis of the design or analysis (adjusted for the most important factor);

f. Comparability of cohorts on the basis of the design or analysis (adjusted for a second important factor);

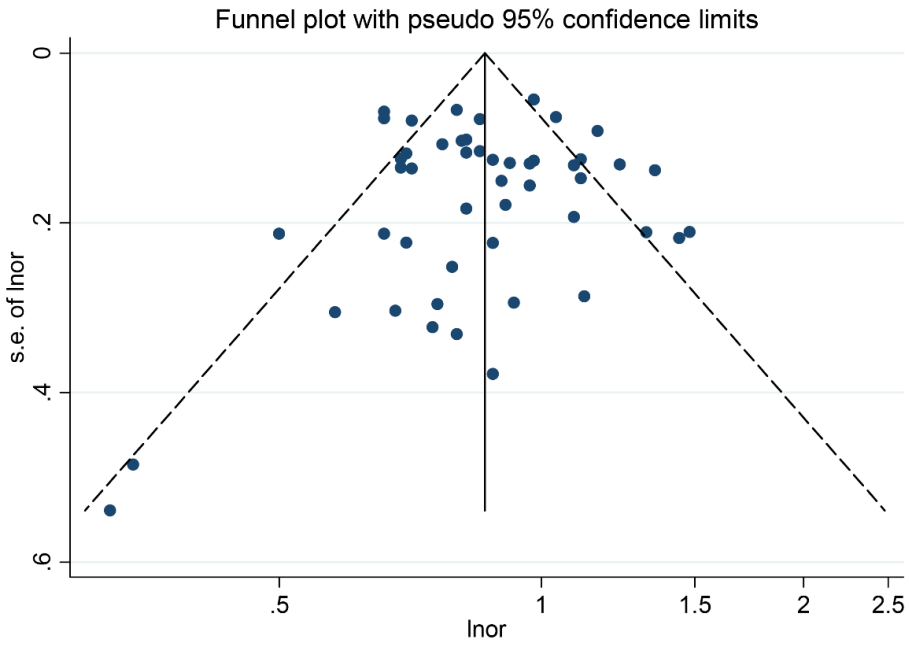
g. Assessment of outcome;

h. Was follow-up long enough for outcomes to occur;

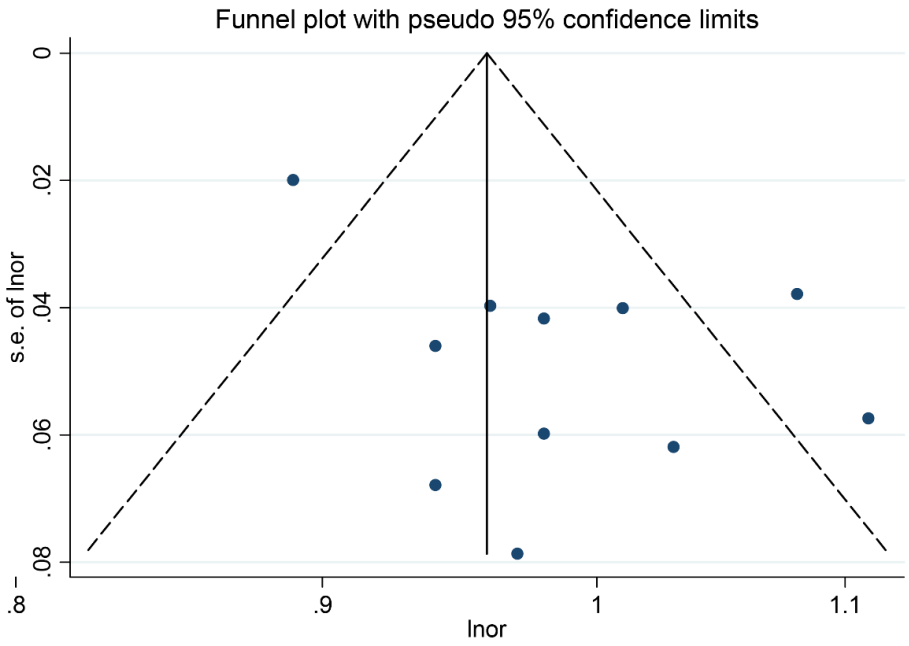
i. Adequacy of follow-up of cohorts.



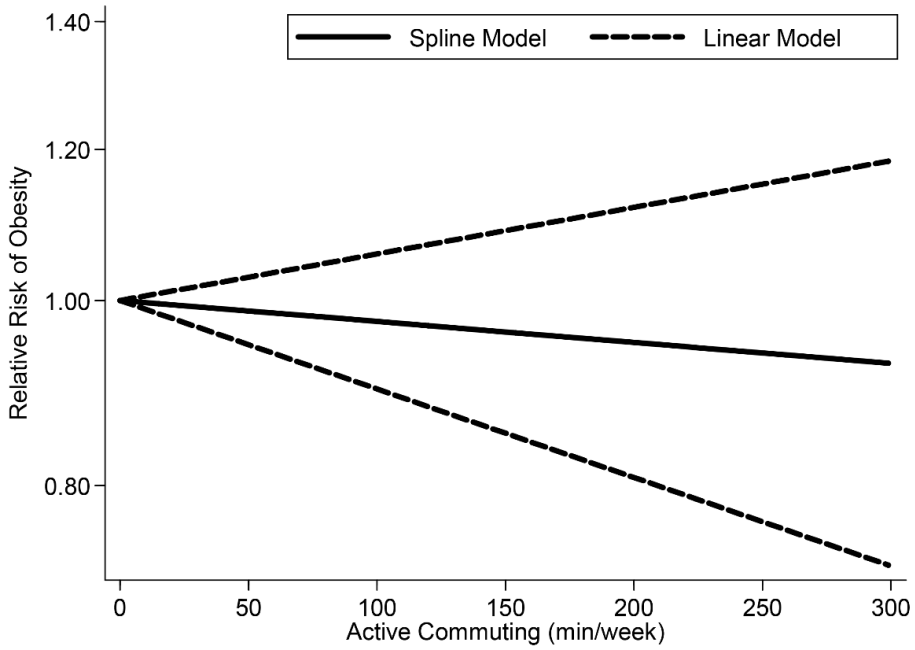
Supplemental Figure 1. Funnel plots for detection of publication bias of included studies for active commuting vs. inactive commuting meta-analysis (active commuting and obesity).



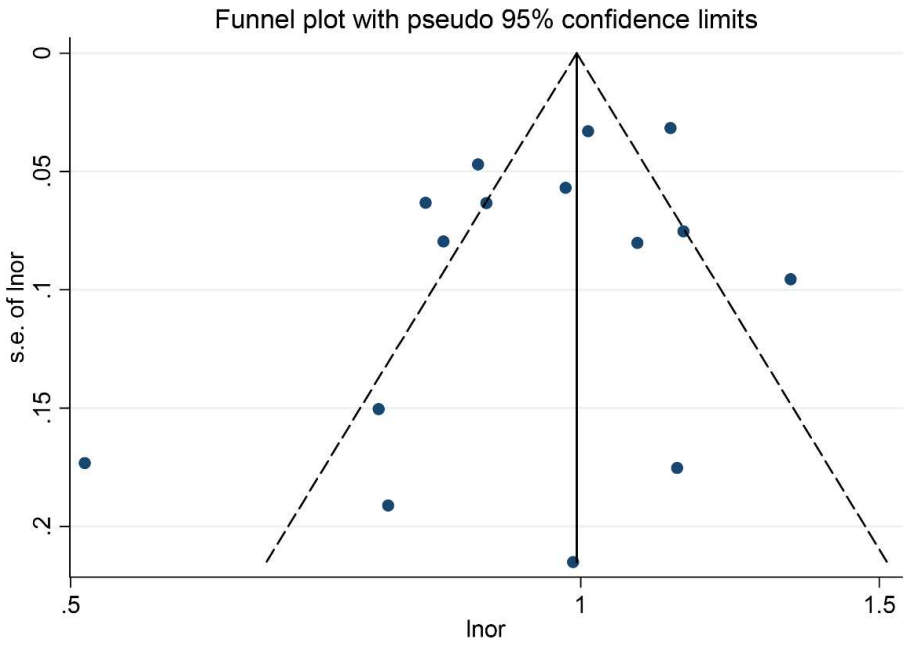
Supplemental Figure 2. Funnel plots for detection of publication bias of included studies for highest vs. lowest active commuting meta-analysis (active commuting and obesity).



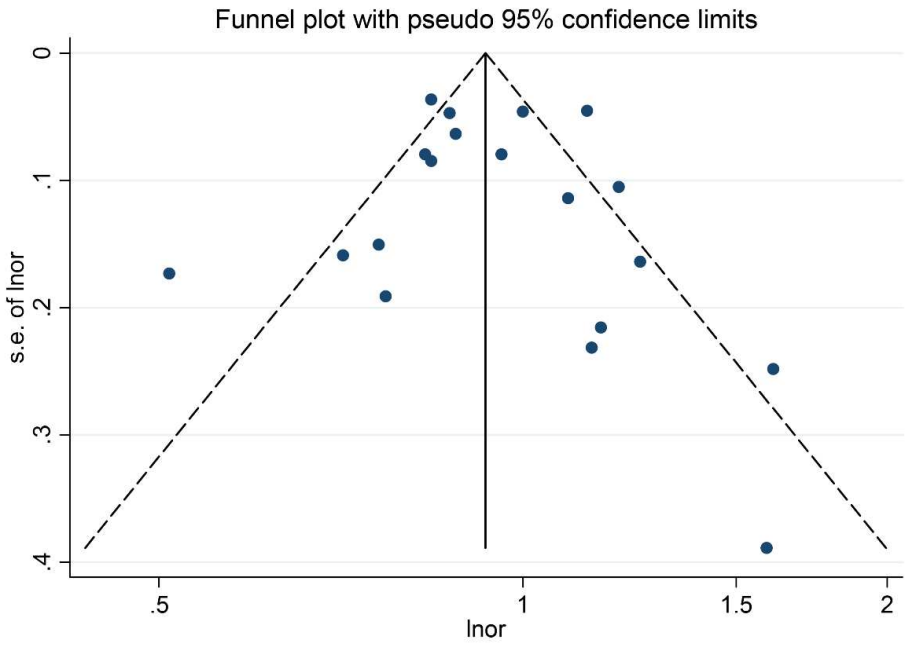
Supplemental Figure 3. Funnel plots for detection of publication bias of included studies for dose–response meta-analysis (active commuting and obesity).



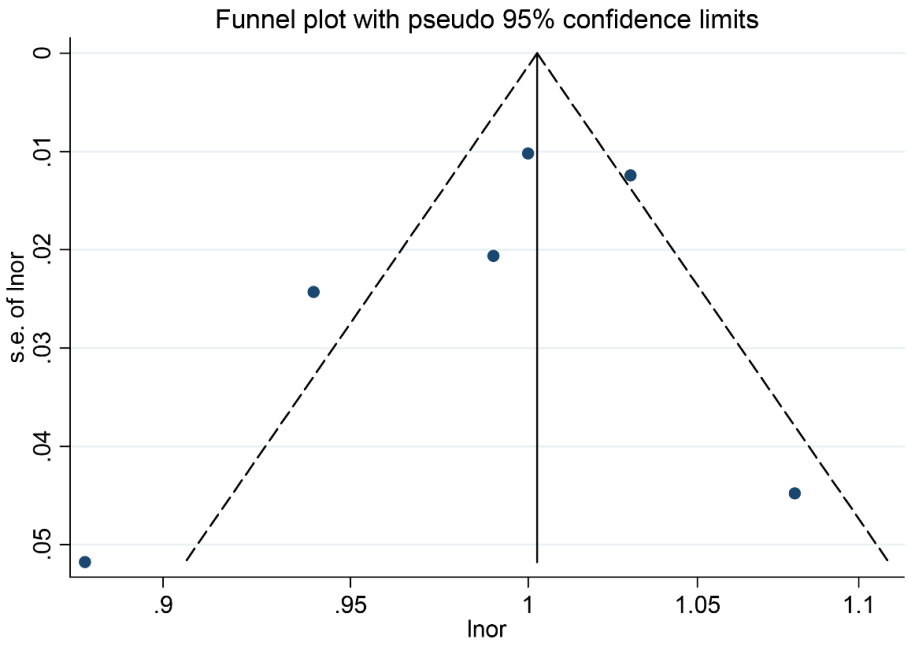
Supplemental Figure 4. Linear dose–response associations between active commuting and obesity modeled by using restricted cubic splines.



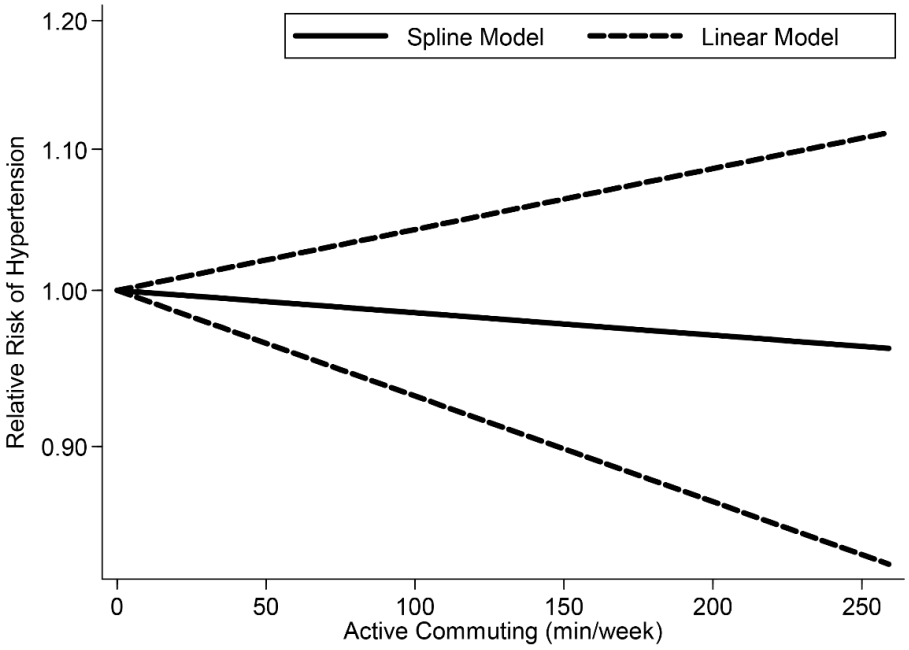
Supplemental Figure 5. Funnel plots for detection of publication bias of included studies for active commuting vs. inactive commuting meta-analysis (active commuting and hypertension).



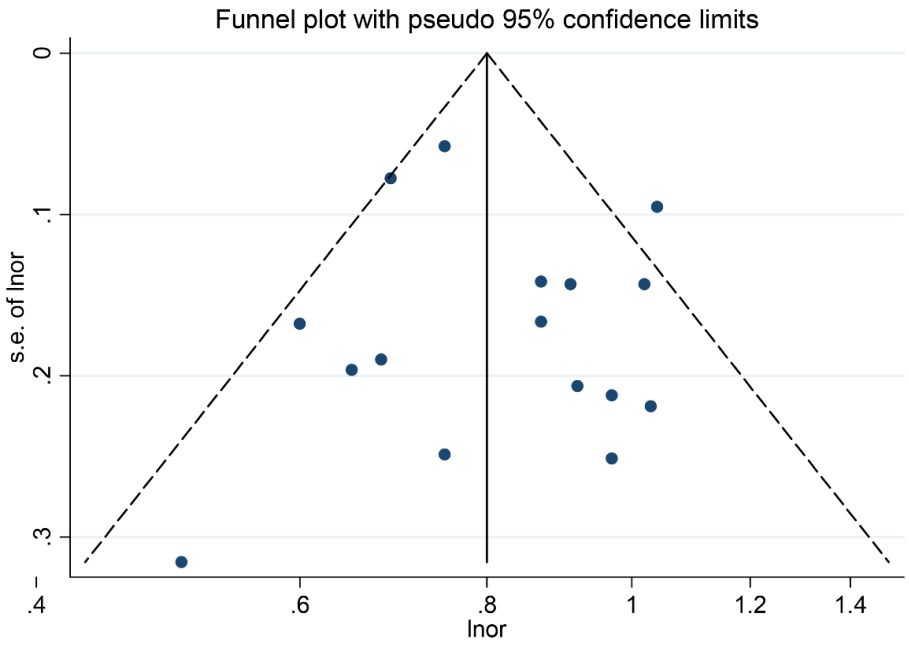
Supplemental Figure 6. Funnel plots for detection of publication bias of included studies for highest vs. lowest active commuting meta-analysis (active commuting and hypertension).



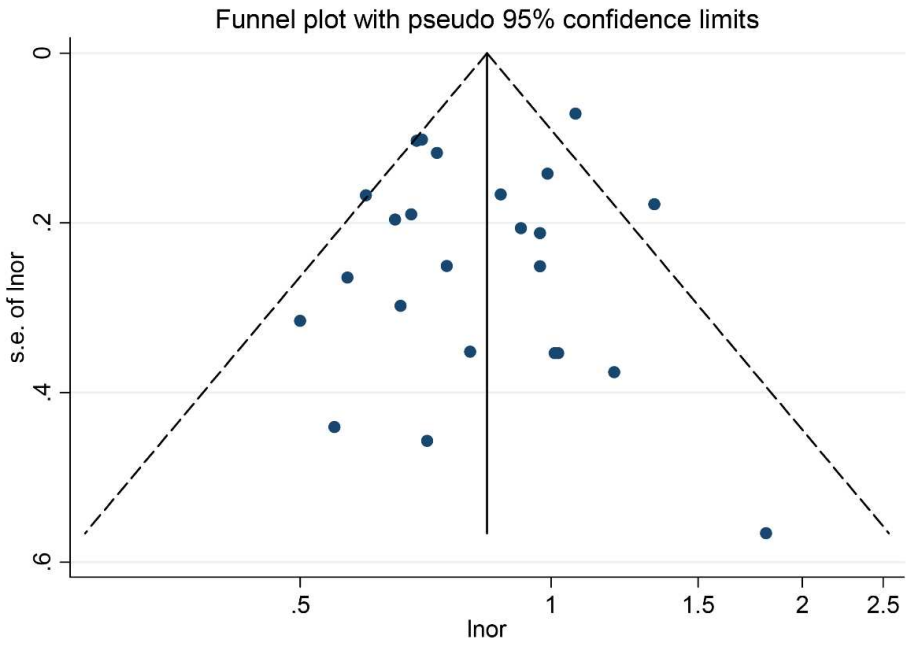
Supplemental Figure 7. Funnel plots for detection of publication bias of included studies for dose–response meta-analysis (active commuting and hypertension).



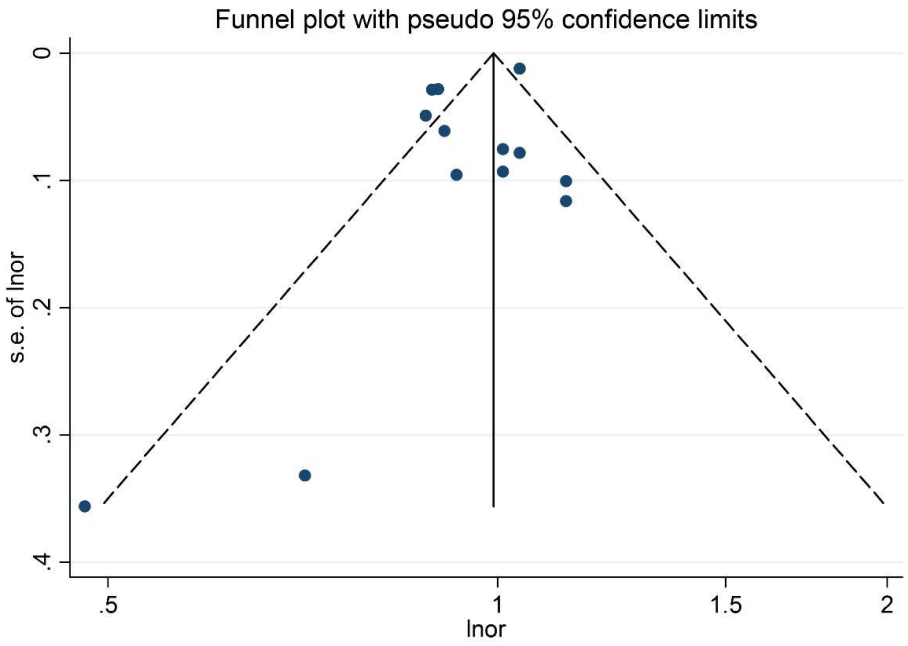
Supplemental Figure 8. Linear dose–response associations between active commuting and hypertension modeled by using restricted cubic splines.



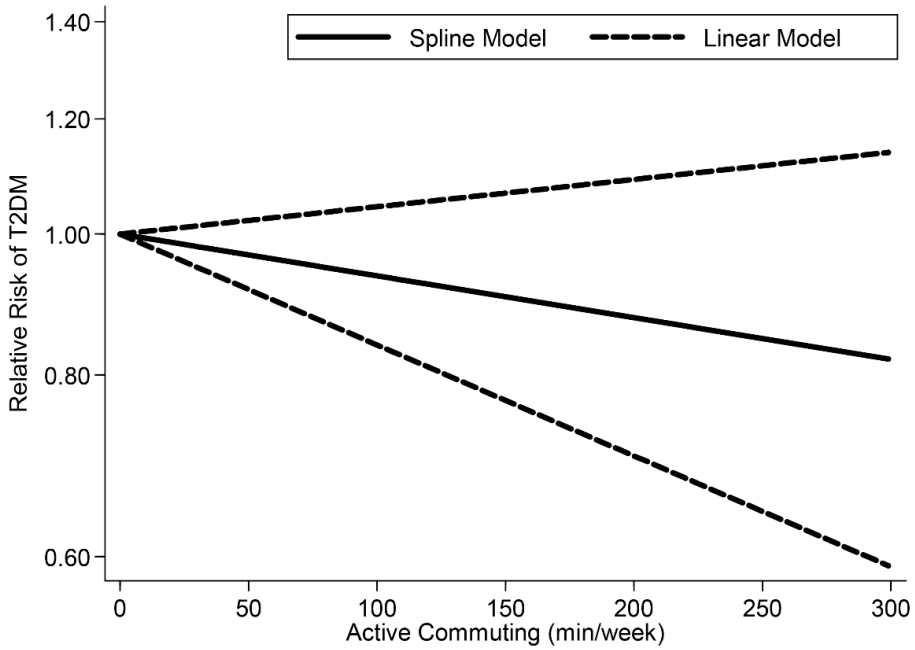
Supplemental Figure 9. Funnel plots for detection of publication bias of included studies for active commuting vs. inactive commuting meta-analysis (active commuting and diabetes).



Supplemental Figure 10. Funnel plots for detection of publication bias of included studies for highest vs. lowest active commuting meta-analysis (active commuting and diabetes).



Supplemental Figure 11. Funnel plots for detection of publication bias of included studies for dose–response meta-analysis (active commuting and diabetes).



Supplemental Figure 12. Linear dose–response associations between active commuting and diabetes modeled by using restricted cubic splines.